

Conway polynomial of the figure-eight knot

We use the oriented skein relation

$$\nabla\left(\begin{array}{c} \nearrow \\ \searrow \end{array}\right) - \nabla\left(\begin{array}{c} \nwarrow \\ \swarrow \end{array}\right) = z \nabla\left(\begin{array}{c} \uparrow \\ \uparrow \end{array}\right), \quad \nabla\left(\begin{array}{c} \circlearrowright \end{array}\right) = 1, \quad \nabla\left(\begin{array}{cc} \circlearrowright & \circlearrowright \end{array}\right) = 0.$$

Choose a crossing of the figure-eight knot. Switching it gives an unknot, while the oriented smoothing gives the negative Hopf link:

$$\begin{aligned} \nabla\left(\begin{array}{c} \text{figure-eight} \end{array}\right) &= \nabla\left(\begin{array}{c} \text{figure-eight} \end{array}\right) + z \nabla\left(\begin{array}{c} \text{Hopf link} \end{array}\right) \\ &= \nabla\left(\begin{array}{c} \text{unknot} \end{array}\right) + z \nabla\left(\begin{array}{c} \text{Hopf link} \end{array}\right). \end{aligned}$$

Apply the skein relation once more to the Hopf link. A crossing change gives the two-component unlink and the smoothing gives an unknot:

$$\begin{aligned} \nabla\left(\begin{array}{c} \text{Hopf link} \end{array}\right) &= \nabla\left(\begin{array}{c} \text{Hopf link} \end{array}\right) - z \nabla\left(\begin{array}{c} \text{unknot} \end{array}\right) \\ &= \nabla\left(\begin{array}{cc} \text{unlink} & \text{unknot} \end{array}\right) - z \nabla\left(\begin{array}{c} \text{unknot} \end{array}\right) \\ &= 0 - z = -z. \end{aligned}$$

Therefore

$$\begin{aligned} \nabla\left(\begin{array}{c} \text{figure-eight} \end{array}\right) &= 1 + z(-z) \\ &= \boxed{1 - z^2}. \end{aligned}$$